Fiscal Year:	FY 2006	**Task Last Updated:**	FY 05/12/2006
PI Name:	Shen, Hayley H Ph.D.		
Project Title:	Constitutive Relation in Transitional Granular Flows		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Granular flows		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	02/13/2002	**End Date:**	12/31/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hanes, Daniel (University of Florida/Hanes and Associates)		
Grant/Contract No.:	NAG3-2717		
Performance Goal No.:			
Performance Goal Text:			
	The behavior of granular materials spans the whole range from an elastic solid to a non-linear viscous fluid. The constitutive relation depends not only on the material composition, but also on the deformation and the rate of deformation. The key reason for this behavior is that granular materials are not only defined by the properties of the individual grains, but also by the structure of the aggregate. In a very dilute granular flow, the structure of the aggregate is simply defined by a uniform random distribution. As the concentration increases, the internal structure becomes complex. It evolves dynamically with the macroscopic deformation and is an integral part of the constitutive relation. The complexity and importance of the internal structure increase with the concentration. In a gravity field, the weight of a granular material naturally compacts it to produce a concentration gradient. To study the constitutive behavior of granular materials, microgravity experiments must be performed to construct a physical basis for transitional constitutive		

Task Description:	relations. For long-term exploration on the Moon and the Mars, rheological properties of the surface granular materials require a better understanding. Moreover, many industrial problems on Earth deal with processing granular materials. Improving fundamental understanding of granular materials will also impact on industrial processes. This project combines both numerical and physical experiments to study the transition of granular flows from solid to gas phases. Using these experiments we will obtain data to quantify the internal resistance of a deforming granular material, to define and measure the internal structures, and to formulate theories to describe the transitional behavior. This study will achieve the following objectives: 1. A “phase diagram” will be established to characterize the transition of granular materials from solid to gas. 2. Physical parameters affecting the transition from one phase to another will be identified. 3. Internal structures of granular flows will be defined and measured. 4. The relation between the internal structures and the transition between phases will be identified. 5. An overall rheological theory for granular materials will be formulated. Flowing granular materials are commonplace in numerous terrestrial processes: landslides, sediment transport, and pneumatic transport of particulate solids in various manufacturing systems. Engineering designs for controlling these flows have relied heavily on empirical data. To date, while humans are capable of launching spacecrafts to land on Mars, failures of grain silos are still an unsolved problem, not to mention large-scale catastrophic granular flows such as debris avalanches. Unlike simple solids or fluids, a granular material under load can display deformation characteristics that cover the entire range from an elastic solid to a variety of non-linear viscous fluids. Many parameters affect the stress, strain, and strain-rate relation: particle size, shape, chemical composition, concentration, non-uniformity, humidity, and even the degree of deformation and rate of deformation. Our study is designed to understand the fundamental mechanisms that govern the rheological behavior of the granular materials so that better engineering designs can be made to solve problems with such materials on Earth as well as other celestial bodies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Improving the fundemental understanding of the rheological properties of granular materials will impact industrial processes such as pharmaceutical powder processing and design of construction (i.e. earth moving) equipment.
Task Progress:	In this year (January – December, 2005) our accomplishments are: 1. Run 3-d code for numerous applications: angle of repose, incline flow, hourglass, penetration tests, simple shear flows. 2. Included electrostatic forces, cluster particles with and without overlaps. 3. Finished building an annular shear cell. 4. Introduced force chain measures to granular flows. Ready for next step of completely mapping the transitional behavior from dilute to dense phase. The Couette shear cell from the Rensselaer Polytechnic Institute was returned to Prof. Kevin Craig. But a copy of it with some minor modifications is built at Clarkson. Currently this annular Couette shear cell is being tested and will be fully functional this year. With this addition Clarkson becomes the only one in the US where all three essential elements in studying granular materials are co-located: theory, computation, and physical experiment. Dr. May Hou from the Chinese Academy of Sciences in Beijing visited Clarkson and Cornell for a week in February 2006. Dr. Hou has a microgravity project to fly on a Chinese satellite to be launched later this year. Her experiment is also invited by the European Space Agency to fly in March of 2006. A paper has finally been accepted from a collaboration initiated by the PI when attending the Microgravity training workshop in Cleveland in 2004. This paper is success story between an engineer and a group of biologists. It is also a success story for having the microgravity workshop that produces interactions between many researchers in different disciplines. Presentations this Year: James Jenkins, Daniel Hanes. Testing DEM using Annular Shear Cell. NASA Lunar/Martian workshop. Feb. 2-3, 2005. Mechanisms and Implications of Pancake Ice Cover Workshop on Modeling of Sea Ice and Ocean Circulation, with an Emphasis on the Sea of Okhotsk. Feb. 18, 2005. Short course: Physics of Granular Flows. Soft Matter Physics Workshop, Chinese Academy of Sciences, Aug. 15-17, 2005.
Bibliography Type:	Description: (Last Updated: 02/13/2008)
Articles in Peer-reviewed Journals	Xie Y, Wang F, Zhong W, Puscheck E, Shen H, Rappolee DA. "Shear Stress Induces Preimplantation Embryo Death That Is Delayed by the Zona Pellucida and Associated with Stress-Activated Protein Kinase-Mediated Apoptosis." Biol Reprod. 2006 Mar 29; [Epub ahead of print] PMID: 16571875 , Mar-2006
Articles in Peer-reviewed Journals	Ji S, Shen HH. "Characteristics of temporal-spatial parameters in quasi-solid-fluid phase transition of granular materials." Chinese Science Bulletin 2006 Mar;51(6):646-54. , Mar-2006
Papers from Meeting Proceedings	Shen HH, Jenkins JT, Ji S. "Geometrically Constrained Flow Behavior with Uniform Disks." The 2005 Joint ASME/ASCE/SES Conference on Mechanics and Materials, Baton Rouge, Louisiana, June 1-3, 2005. Proc. McMat2005, Paper No. 624, June 2005. , Jun-2005